Differentiation & Integration

ANSWER KEY

Section 1: Derivatives of Polynomials & Tangents

1 Differentiate $y = 3x^4 - 5x^2 + 7x - 2$ with respect to x.

a. $12x^3 - 10x + 7$ ✓

b. $12x^3 - 10x^2 + 7$

c. $7x^3 - 10x + 7$

d. $12x^4 - 10x^2 + 7$

Apply the power rule: $d/dx[3x^4] = 12x^3$, $d/dx[-5x^2] = -10x$, $d/dx[7x] = 7$, $d/dx[-2] = 0$.

2 Find the gradient of the curve $y = x^3 - 4x + 1$ at the point where $x = 2$.

a. **8** ✓

b. 12

c. 4

d. 0

$dy/dx = 3x^2 - 4$. At $x = 2$: $dy/dx = 3(4) - 4 = 12 - 4 = 8$.

3 Find the equation of the tangent to the curve $y = x^2 - 3x + 2$ at the point $(3, 2)$.

a. $y = 3x - 7$ ✓

b. $y = 3x - 2$

c. $y = x - 1$

d. $y = 2x - 4$

$dy/dx = 2x - 3$. At $x = 3$, gradient $m = 2(3) - 3 = 3$. Tangent: $y - 2 = 3(x - 3) \Rightarrow y = 3x - 7$.

Section 2: Stationary Points & Optimization

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Find the stationary points of the curve $y = 2x^3 - 9x^2 + 12x + 1$ and determine their nature.

a. **Local max at $(1, 6)$, local min at $(2, 5)$** ✓

b. Local min at $(1, 6)$, local max at $(2, 5)$

c. Local max at $(2, 5)$, inflection at $(1, 6)$

d. Local min at $(0, 1)$, local max at $(2, 5)$

$dy/dx = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2) = 0 \Rightarrow x = 1, 2$. $d^2y/dx^2 = 12x - 18$. At $x = 1$: $d^2y/dx^2 = -6 < 0$ (max, $y = 6$). At $x = 2$: $d^2y/dx^2 = 6 > 0$ (min, $y = 5$).

5 A metal wire of length 40 cm is bent into a rectangle of dimensions x by y. What value of x maximizes the enclosed area?

a. $x = 10$ cm ✓

b. $x = 20$ cm

c. $x = 5$ cm

d. $x = 15$ cm

$2x + 2y = 40 \Rightarrow y = 20 - x$. Area $A = x(20 - x) = 20x - x^2$. $dA/dx = 20 - 2x = 0 \Rightarrow x = 10$.

6 For what range of values of x is the function $f(x) = x^3 - 6x^2 + 9x - 4$ strictly increasing?

a. $x < 1$ **or** $x > 3$ ✓

b. $1 < x < 3$

c. $x > 2$

d. $x < 2$

$f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3) > 0 \Rightarrow x < 1$ or $x > 3$.

Section 3: Indefinite & Definite Integrals

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Find $\int (6x^2 - 4x + 5)\,dx$.

a. $2x^3 - 2x^2 + 5x + C$ ✓

b. $3x^3 - 2x^2 + 5x + C$

c. $2x^3 - 4x^2 + 5x + C$

d. $12x - 4 + C$

$\int 6x^2 dx = 2x^3$, $\int -4x dx = -2x^2$, $\int 5 dx = 5x$. Result: $2x^3 - 2x^2 + 5x + C$.

8 Evaluate the definite integral $\int_1^3 (3x^2 - 2x)\,dx$.

a. **18** ✓

b. 24

c. 16

d. 20

$[x^3 - x^2]_1^3 = (27 - 9) - (1 - 1) = 18 - 0 = 18$.

9 A curve passes through $(1, 4)$ and has gradient function $\frac{dy}{dx} = 4x - 1$. Find the equation of the curve.

a. $y = 2x^2 - x + 3$ ✓

b. $y = 2x^2 - x + 4$

c. $y = 4x^2 - x + 1$

d. $y = 2x^2 - x + 1$

$y = \int (4x - 1) dx = 2x^2 - x + C$. At $(1, 4)$: $4 = 2(1) - 1 + C \Rightarrow 4 = 1 + C \Rightarrow C = 3$. Curve: $y = 2x^2 - x + 3$.

Section 4: Area Under Curves & Exam-Style Modeling

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Find the area enclosed by the curve $y = 4 - x^2$ and the x-axis.

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 8

d. $\frac{64}{3}$

Roots at $x = \pm 2$. Area
$= \int_{-2}^{2}(4 - x^2)dx = [4x - x^3/3]_{-2}^{2} = (8 - 8/3) - (-8 + 8/3) = 16/3 - (-16/3) = 32/3.$

11 Find the area of the region bounded by the curve $y = x^2$ and the line $y = 2x$.

a. $\frac{4}{3}$ ✓

b. $\frac{2}{3}$

c. 2

d. $\frac{8}{3}$

Intersections: $x^2 = 2x \Rightarrow x(x - 2) = 0 \Rightarrow x = 0, 2$. Area
$= \int_{0}^{2}(2x - x^2)dx = [x^2 - x^3/3]_{0}^{2} = 4 - 8/3 = 4/3.$

12 A particle moves along a straight line such that its velocity at time t is $v(t) = 3t^2 - 12t + 9$. Find the total displacement between $t = 0$ and $t = 3$.

a. **0** ✓

b. 9

c. 18

d. 4.5

Displacement $= \int_{0}^{3}(3t^2 - 12t + 9)dt = [t^3 - 6t^2 + 9t]_{0}^{3} = (27 - 54 + 27) - 0 = 0.$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647